WORKED SOLUTION

STEP II 2003, Question 11

Mechanics • Projectile Motion

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Read the question before you go on. It is in the STEP Support Programme's question database:

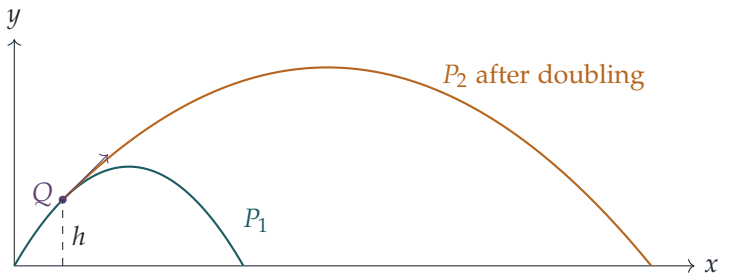
step.maths.org/questions/03-s2-q11

The key idea

Work with the horizontal and vertical components of velocity separately. The direction of motion makes 45° with the horizontal exactly when the two components are equal, and that gives the first time directly. When the speed is doubled the direction doesn't change, so both components double, and the rest of the flight is a new projectile problem that starts at a known height.

Solution

We model each particle as a point projectile moving under gravity alone, with no air resistance. Write $s = \sin\alpha$ and $c = \cos\alpha$. Since $\alpha > 45^\circ$ we have $s > c > 0$.



The sketch uses $\alpha = 60^\circ$. Both particles follow the teal path as far as Q, where the direction first makes 45° with the horizontal. After that P_2 follows the amber path.

The flight time of P_1

The vertical displacement at time t is $y = Vs\,t - \frac{1}{2}gt^2 = t\left(Vs - \frac{1}{2}gt\right)$. This is zero again when

$$T_1 = \frac{2V\sin\alpha}{g}.$$

The first 45° moment

At time t the velocity components are

$$\dot{x} = Vc, \qquad \dot{y} = Vs - gt.$$

The horizontal component never changes. At launch the direction is steeper than 45°, because $\dot{y} = Vs > Vc = \dot{x}$. As $\dot{y}$ decreases, the angle first falls to 45° when $\dot{y} = \dot{x}$, with the particle still rising:

$$Vs - gt_1 = Vc \quad \Longrightarrow \quad t_1 = \frac{V(s-c)}{g}.$$

(The direction makes 45° again on the way down, when $\dot{y} = -Vc$, but that is later.) Since $\frac{1}{2}T_1 = Vs/g$,

$$t_1 = \frac{Vs}{g}\left(1 - \frac{c}{s}\right) = \tfrac{1}{2}(1 - \cot\alpha)\, T_1,$$

as required.

The flight of P_2

Up to time t_1, P_2 moves exactly like P_1. At t_1 both velocity components equal Vc, and the height is

$$h = Vs\,t_1 - \tfrac{1}{2}gt_1^2 = \frac{V^2}{g}\left[s(s-c) - \tfrac{1}{2}(s-c)^2\right]$$
$$= \frac{V^2}{2g}(s-c)[2s - (s-c)] = \frac{V^2}{2g}\,(s^2 - c^2).$$

This is positive because $s > c$.

Ask yourself What does "the speed is doubled" do to each component of the velocity?

The speed changes but the direction doesn't, so the whole velocity vector is multiplied by 2. Both components become $2Vc$: the particle sets off from height h at 45° above the horizontal with speed $2\sqrt{2}\,Vc$.

Let τ be the time from the doubling until P_2 lands. Measuring vertical displacement upwards from Q, landing means a displacement of $-h$:

$$2Vc\,\tau - \tfrac{1}{2}g\tau^2 = -\frac{V^2}{2g}(s^2 - c^2).$$

Multiply by $-2/g$ and write $\tau = (V/g)u$, where u is dimensionless:

$$u^2 - 4cu - (s^2 - c^2) = 0 \quad \Longrightarrow \quad u = 2c \pm \sqrt{4c^2 + s^2 - c^2} = 2c \pm \sqrt{s^2 + 3c^2}.$$

The product of the roots is $-(s^2 - c^2) < 0$, so one root is negative. We need $\tau > 0$, so we take the plus sign:

$$\tau = \frac{V}{g}\left(2c + \sqrt{s^2 + 3c^2}\right).$$

Therefore

$$T_2 = t_1 + \tau = \frac{V}{g}\left(s - c + 2c + \sqrt{s^2 + 3c^2}\right) = \frac{V}{g}\left(s + c + \sqrt{s^2 + 3c^2}\right).$$

Dividing by $\frac{1}{2}T_1 = Vs/g$ and using $\sqrt{s^2 + 3c^2}/s = \sqrt{1 + 3c^2/s^2}$ (valid because $s > 0$):

$$\frac{2T_2}{T_1} = \frac{s + c + \sqrt{s^2 + 3c^2}}{s} = 1 + \cot\alpha + \sqrt{1 + 3\cot^2\alpha},$$

which is the required result.

Checks

- As $\alpha \to 90^\circ$, $\cot\alpha \to 0$ and the ratio tends to 2. This makes sense: the 45° moment is now the top of the flight, where the speed is zero, so doubling it does nothing and $T_2 = T_1$.
- At $\alpha = 45^\circ$ (the boundary case) the doubling happens at launch, so P_2 is just projected at $2V$, and $T_2 = 2T_1$. The formula gives $1 + 1 + 2 = 4$, which agrees.
- At $\alpha = 60^\circ$, $t_1/T_1 = \frac{1}{2}(1 - 1/\sqrt{3}) \approx 0.2113$ and $2T_2/T_1 = 1 + 1/\sqrt{3} + \sqrt{2} \approx 2.9916$. A numerical step-by-step simulation of P_2's flight for 2000 random choices of α and V matched the formula for T_2.

Common slips

- Doubling only the vertical component, or only the horizontal one. The direction is unchanged, so both double.
- Restarting P_2's motion from ground level instead of from height h.
- Taking the negative root for τ without checking its sign.

Beyond STEP

An instantaneous change of velocity is an *impulse*: a very large force acting for a very short time. Impulses are how first-year mechanics treats collisions, rocket burns and kicks. Splitting a trajectory into pieces joined by impulses is exactly how orbital manoeuvres are planned, where a short engine burn moves a spacecraft from one conic orbit onto another. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops impulse and momentum carefully and has many problems of this kind.

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